

FINAL YEAR PROJECT — PFE

ESPRIT · 2026

PEAXIS

An AI-Powered Hiring Operating System

BIBANI Mohamed Habib Allah

École Supérieure Privée d'Ingénierie et de Technologie — ESPRIT

• Full-Stack Engineer · AI Integration · Platform Architect

Next.js

NestJS

FastAPI

PostgreSQL

Gemini

Docker

What is PEAXIS?

A modular SaaS recruitment platform that combines AI-powered hiring intelligence with a best-in-class candidate experience — built from scratch during this PFE.

MISSION

Eliminate hiring friction for growing companies by making every recruitment decision faster, smarter, and explainable.



PEAXIS Core

NestJS REST API — central data authority, auth, billing, multi-tenant isolation



PEAXIS Hire

Next.js recruiter dashboard — pipeline Kanban, AI insights, analytics



PEAXIS Jobs

Next.js candidate portal — job discovery, CV upload, application tracking



AI Brain

FastAPI service — CV parsing, match scoring, hiring copilot, JD generation

HR TECH

SAAS B2B

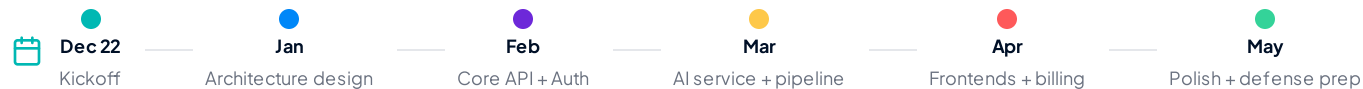
AI/ML

RECRUITMENT AUTOMATION

2 INTERNSHIP CONTEXT

Building PEAXIS from **day zero**

A 6-month final year internship starting December 22 — sole engineer responsible for design, architecture, and full implementation.



Full-Stack Architecture

- NestJS API (14 modules)
- Next.js Hire + Jobs portals
- Multi-tenant data model
- BullMQ background pipeline



AI Integration

- FastAPI AI microservice
- CV parsing (Gemini)
- Match scoring engine
- Hiring Copilot
- JD Generator



Platform Engineering

- Docker dev + prod setup
- Prisma schema + migrations
- pgvector semantic search
- Stripe billing integration

FULL-STACK ENGINEER

AI INTEGRATION

PLATFORM ARCHITECT

SOLO CONTRIBUTOR

3 THE PROBLEM

Recruitment is **broken** at every layer

The hiring process hasn't fundamentally changed in 20 years. Time, money, and talent are wasted at every step.

42 days

Average time-to-hire

SHRM 2024

75%

Recruiters overwhelmed by CV volume

LinkedIn Talent Report

60%

Candidates ghost after poor UX

Indeed Survey 2023



Screening bottleneck

Recruiters spend 23 hours per week manually reviewing CVs. 80% of that time is wasted on unqualified applications.



Bias & inconsistency

Manual screening introduces unconscious bias and inconsistent evaluation criteria across reviewers and hiring rounds.



Candidate friction

Long applications, no feedback, and opaque processes cause top talent to abandon their candidacies before completion.

The result: companies lose an average of **\$14,900** per bad hire, while top candidates accept competing offers in less than 10 days.

3 THE PROBLEM

Legacy ATS tools are **not enough**

Traditional Applicant Tracking Systems were built to store applications — not to intelligently process them.

CAPABILITY	LEGACY ATS	PEAXIS
AI-powered candidate ranking	✗	✓
Explainable match scores	✗	✓
Automatic CV parsing	✗	✓
Candidate-facing portal	basic	✓
Real-time analytics pipeline	✗	✓
Semantic job search	✗	✓
Hiring copilot / AI assistant	✗	✓
Multi-tenant SaaS architecture	partial	✓

The market gap: intelligent + integrated + explainable



AI black boxes

"This candidate scored 72" — but why? Recruiters cannot act on scores they cannot explain. Bias complaints follow.



Fragmented tooling

Notion for job specs. Greenhouse for pipeline. Calendly for interviews. Zero integration — context lost at every handoff.



No candidate intelligence

Existing tools track applications. None actively guide recruiters on who to advance, how to interview, or what risks to flag.

\$1.2B

HR Tech VC funding in 2024

23%

CAGR of AI Recruitment market

78%

HR leaders plan AI adoption in 2025



PEAXIS fills this exact gap

AI scores are explainable. Tools are integrated. The candidate experience is first-class. That's the OS layer missing from the market.

4 FUNCTIONAL REQUIREMENTS

Core platform requirements

FR-01 through FR-05 cover the foundational platform capabilities required for a production-grade SaaS recruitment system.



FR-01

Authentication & Security

- Email/password with Argon2 hashing
- OAuth2: Google + LinkedIn
- JWT access (15m) + refresh (7d)
- Rate limiting: 1000 req/60s



FR-02

Multi-Tenant Architecture

- Business-scoped data isolation
- X-Business-ID header enforcement
- User ↔ Business join table
- Unique business slug routing



FR-03

Roles & Permissions

- Plan-based feature gating
- Free / Starter / Pro / Enterprise tiers
- JwtAuthGuard + PlanGuard
- Business member roles



FR-04

Job Management

- Full CRUD with rich filters
- Semantic search via pgvector
- Auto embedding generation
- Type, location, salary, skills filters



FR-05

Application Pipeline

- 6-stage:
Applied → Screening → Interview → Offer → Hired/Rejected
- Stage transition history
- AI scoring at application time
- Recruiter kanban view

4 FUNCTIONAL REQUIREMENTS

AI & analytics requirements

FR-06 through FR-10 define the intelligence layer — the core differentiator that elevates PEAXIS above traditional ATS tools.



FR-06

AI Match Scoring

- Deterministic skill-overlap engine
- Score: 0–100 with explanation
- Strengths & weaknesses breakdown
- Applied at submission time



FR-07

CV Parsing

- PDF / DOCX / TXT support
- Gemini 2.5–flash extraction
- Structured JSON output
- Auto pre-fills candidate profile



FR-08

Candidate Onboarding

- Anonymous apply-with-CV flow
- Profile completeness scoring
- Semantic embedding generation
- Email verification



FR-09

Interview Scheduling

- Interview CRUD with type flags
- Linked to application record
- Interviewer assignment
- Status tracking



FR-10

Analytics Dashboard

- Pipeline funnel metrics
- Score distribution charts
- Time-to-hire tracking
- Per-job + global analytics

Production-grade **quality attributes**



Horizontal scale

NFR-01 **Scalability**

Stateless API pods behind load balancer. BullMQ handles surge traffic via Redis queue. pgvector scales to millions of embeddings.



OWASP aligned

NFR-02 **Security**

Argon2 passwords, JWT rotation, OAuth2, X-Business-ID multi-tenant isolation, OWASP-hardened API, X-Service-Secret inter-service auth.



p95 < 200ms

NFR-03 **Performance**

Redis 3-layer caching (embeddings 7d, CV parse 30d, AI results 24h). Sub-200ms API p95. Async BullMQ for heavy operations.



99.5% target SLA

NFR-04 **Availability**

Preferred-region retries in SDK. BullMQ retry logic. Docker health checks. Graceful degradation on AI service downtime.



Full audit trail

NFR-05 **Explainability**

Every AI score exposes: numeric value, label (Strong/Good/Weak), matched skills, missing skills, and a plain-language explanation.



Microservices

NFR-06 **Modularity**

NestJS feature modules. Independent AI service. Decoupled frontend apps. BullMQ workers isolated from request handlers.



Cache-first AI

NFR-07 **Cost Optimization**

Gemini Flash for fast tasks. GPT-4.1-mini for quality outputs. Redis caching prevents redundant LLM calls. Docker resource limits.

PEAXIS — The Hiring Operating System

Four deeply integrated products that replace the fragmented stack every recruiter tolerates today.



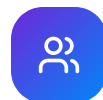
API Layer
PEAXIS Core

NestJS REST API — single source of truth. Auth, billing, multi-tenant, all data owned here.



Recruiter Dashboard
PEAXIS Hire

Next.js SPA — pipeline Kanban, AI insights panel, job management, analytics.



Candidate Portal
PEAXIS Jobs

Next.js SPA — semantic job search, CV upload & parse, application tracking.



FastAPI Service
AI Brain

Stateless Python service — CV parsing, match scoring, copilot, JD generation.

● All four products share a single Prisma → PostgreSQL data layer

NEXT.JS 16

NESTJS

FASTAPI

POSTGRESQL + PGVECTOR

REDIS

BULLMQ

GEMINI

STRIPE

6 SOLUTION

PEAXIS Hire — Recruiter Dashboard NEXT.JS 16

The workspace where recruiters live. Every hiring decision powered by AI, every action one click away.



Pipeline Kanban

Drag-and-drop stage management with AI scores visible on every card



AI Insight Panel

Match score, matched/missing skills, AI explanation per candidate



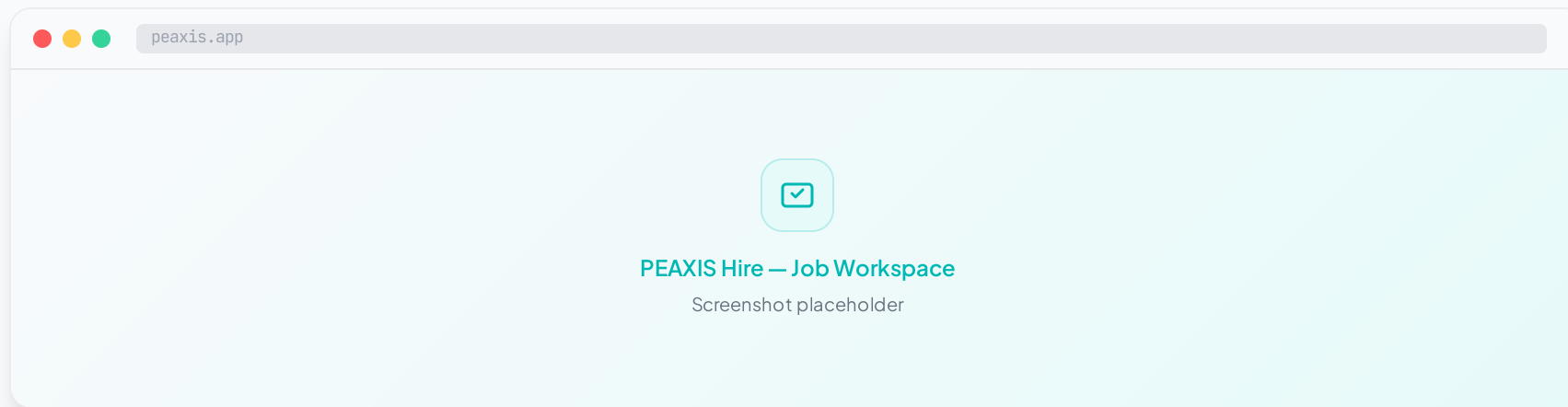
Job Analytics

Funnel metrics, score distributions, time-to-stage per job



Inline Editing

Edit job details inline without leaving the workspace — no navigation



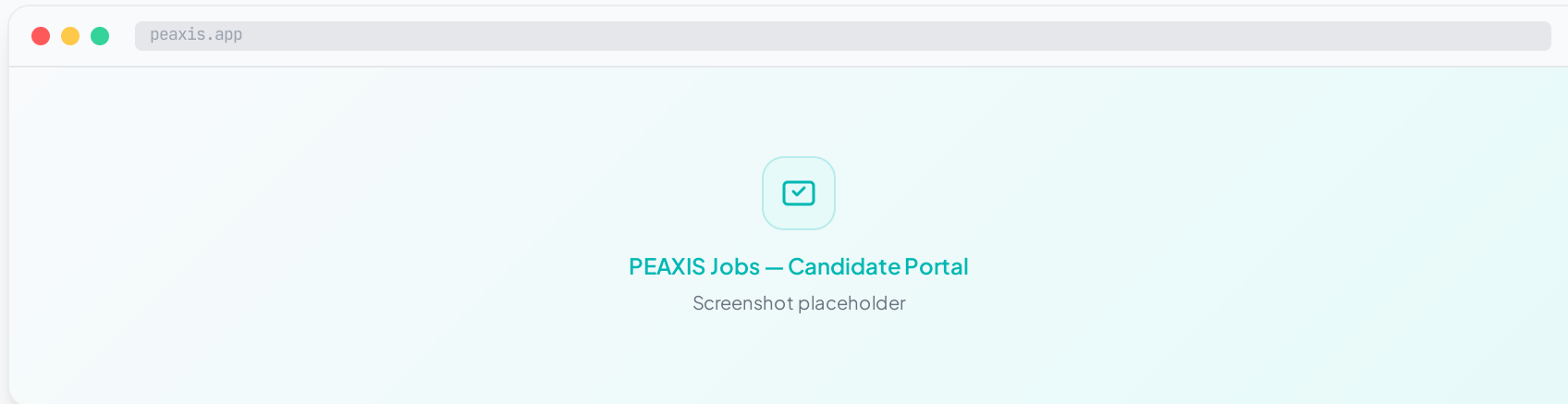
Job Workspace: Pipeline Kanban + Candidate Drawer + AI Insights

Architecture: Server components fetch all data in parallel (Promise.all). Client components own interactive state only. Candidate card click opens a slide-in **CandidateDrawer** — never navigates away.

PEAXIS Jobs — Candidate Portal

NEXT.JS 16

The candidate experience that makes applying frictionless — from discovery to tracking.



Candidate Portal: Job Feed + CV Upload + Application Tracker



Semantic Job Search

pgvector-powered search — finds relevant jobs even with different wording



Apply with CV

Upload CV → AI parses it → profile pre-filled → one-click apply



Application Tracker

Real-time pipeline stage visibility — Applied, Screening, Interview, Offer



Job Recommendations

Personalised job feed powered by candidate embedding similarity

Upload CV (PDF/DOCX) → AI parses (Gemini) → Profile pre-filled → One-click apply → Real-time status

7 TECHNOLOGIES

Frontend & API stack

Frontend Layer

PEAXIS Hire

Recruiter dashboard — port 3000, Bun runtime

PEAXIS Jobs

Candidate portal — port 3001, Bun runtime

Rendering

Server components fetch → client components own state

Next.js

React

Tailwind

TypeScript

API Layer

Core API

14 modules — port 4000, single source of truth

Auth

JWT RS256 + Passport OAuth2 (Google, LinkedIn)

Queue

BullMQ workers for embeddings + AI scoring

NestJS

TypeScript

Prisma

JWT

Database Layer

PostgreSQL

Primary datastore — all persistent data

pgvector

1536-dim embeddings for semantic search

Redis

Cache + BullMQ queue backend

PostgreSQL

pgvector

Prisma

Password hashing
Argon2id

Session tokens
JWT HS256 • 15m / 7d

Rate limiting
1000 req / 60s global

AI & infrastructure stack

AI FEATURE	MODEL	WHY
● CV Parsing	Gemini 2.5-flash	Fast structured JSON, low cost
● Match Scoring	Deterministic	Reproducible, explainable, no LLM needed
● JD Generation	Gemini 2.5-flash	Structured output + bias checking
● Hiring Copilot	Gemini 2.5-flash	Summaries, risk flags, interview Qs
● Cover Letters	GPT-4.1-mini	Higher creative quality
● Embeddings	text-embedding-3-small 1536-dim, cost-effective	

Infrastructure

FastAPI AI service — stateless, Python 3.12+, Pydantic schemas

Redis 3-layer cache: embeddings 7d · CV parse 30d · results 24h

BullMQ Background queue for embeddings, scoring, analysis jobs

Docker Dev + prod compose files, multi-stage Dockerfiles

Stripe Subscription billing with webhook event handling

Redis Cache-first strategy: sha256(model+input) key → hit avoids LLM call entirely. Embeddings cached 7 days, CV parses 30 days, general AI results 24 hours.

8 LOGICAL ARCHITECTURE

System layered architecture

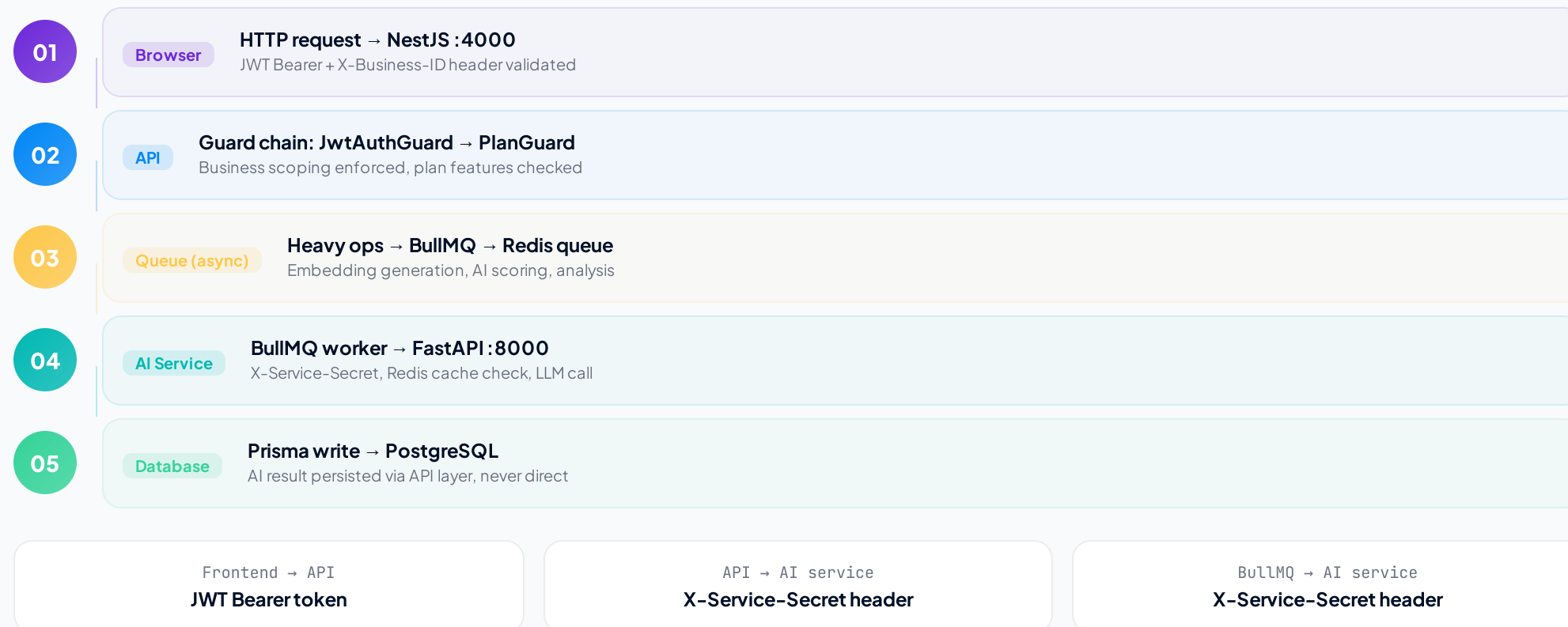
Each layer has a single, clear responsibility. No layer bypasses the one above it.



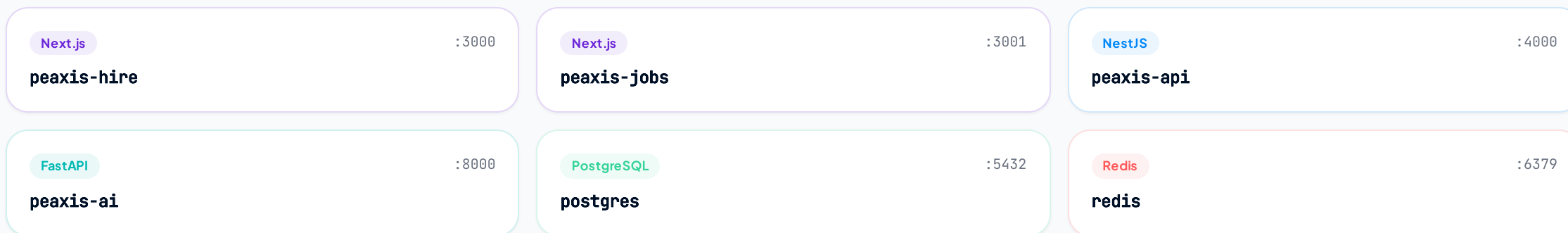
Key principle: The NestJS API is the *single source of truth*. The AI service never writes to the database — it returns results, the API persists them via Prisma.

Request lifecycle & data flow

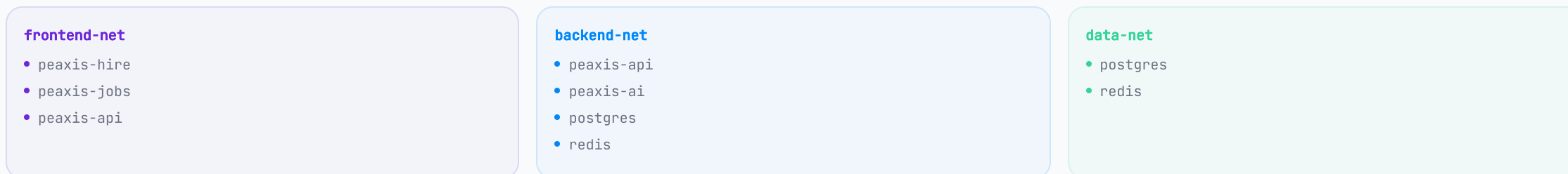
From browser to database — every hop is authenticated, scoped, and auditable.



Docker Compose deployment topology



DOCKER NETWORKS



Exposed to host: :3000 (Hire) :3001 (Jobs) :4000 (API) :8000 (AI) :5432 (PG dev) :6379 (Redis dev)

Infrastructure & DevOps setup

Development

`docker-compose.yml`

Infra only (Postgres, Redis). Frontend/API run natively with hot-reload.

- postgres:5432
- redis:6379
- API: native 4000
- AI: native 8000

Production

`docker-compose.prod.yml`

All services containerised with multi-stage Dockerfiles and resource limits.

- All 6 services
- Multi-stage builds
- Resource limits
- Health checks

🔒 Secrets management

All credentials via environment variables. `.env.example` committed. No secrets in version control.

⚙️ Config via ConfigService

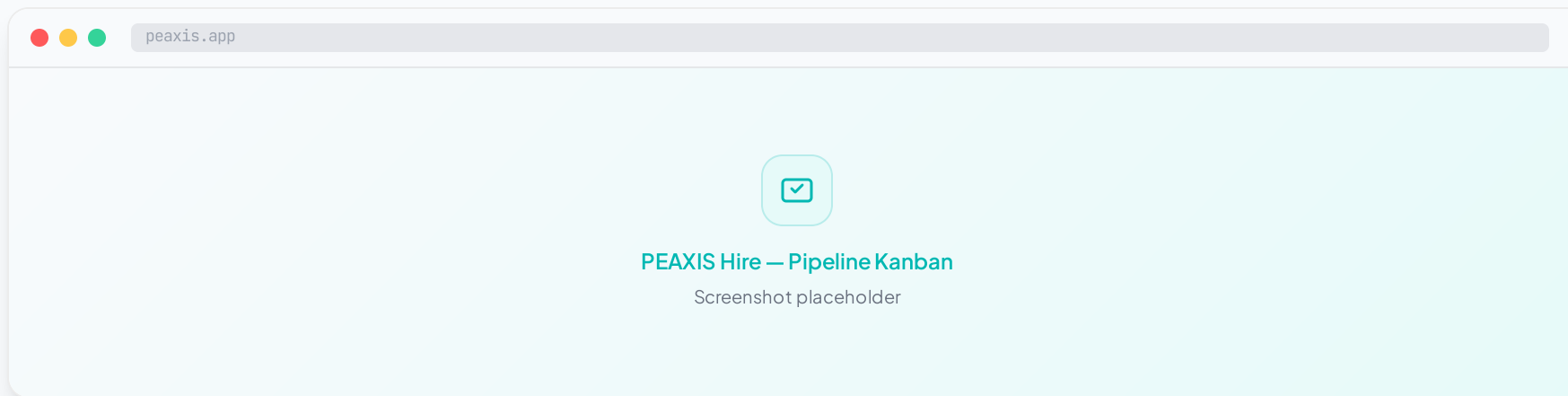
NestJS ConfigService wraps `process.env`. Direct `process.env` access is prohibited in application code.

🔄 Prisma migrations

Schema-first with `prisma migrate dev`. All migrations version-controlled in `prisma/migrations/`.

Observability: NestJS Logger / HumanysLogger for structured logs. No `console.log`. FastAPI logs routed via `uvicorn`. Redis diagnostic string captured on latency spikes.

Demo: Recruiter Dashboard



Pipeline view — drag cards between stages, click to open Candidate Drawer

Job Workspace

All job-related data in one tabbed view
— no page hops

Pipeline Kanban

Drag-and-drop stage management
with AI scores on every card

Candidate Drawer

Slide-in panel reveals AI insights without
leaving the pipeline

Inline Job Edit

Edit job details within the workspace —
router.refresh() on save

UX principle: Candidate card click *never navigates* — it opens the CandidateDrawer (slide-in panel). "View Full Profile" inside the drawer navigates. One action, clear intent.

Demo: Candidate Flow

From job discovery to application tracking — designed to feel effortless.



1. Discover

Candidate finds job via semantic search



2. Upload CV

PDF/DOCX uploaded to public endpoint



3. AI Parse

Gemini extracts structured profile data



4. Pre-fill

Application form auto-filled from CV



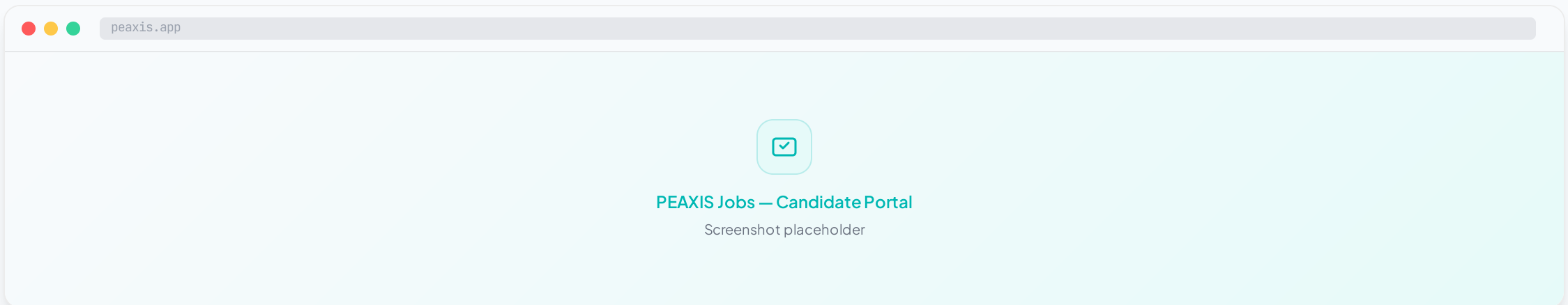
5. Apply

Application created + AI score computed



6. Track

Real-time pipeline stage visibility



Job Feed + Application Tracker — candidate sees real-time pipeline progress

CV parsing: sha256(model + text) cache key → Redis (30d TTL). Identical CVs never re-parsed. Gemini 2.5-flash returns `name`, `email`, `phone`, `skills[]`, `experience[]`, `education[]`, `summary`.

Demo: AI Engine

● Match Scoring

Deterministic skill-overlap → 0–100 score with matched/missing skills + plain-language explanation

● CV Parsing

Gemini 2.5–flash → structured JSON profile from PDF/DOCX/TXT in under 3 seconds

● Hiring Copilot

4 functions: candidate summary, interview questions, risk flags, suggest next action

● JD Generator

Structured job description with bias-checking prompt and optional skill suggestions

★ AI Match Score

82 /100

Strong Match

React ✓

TypeScript ✓

Node.js ✓

GraphQL ✗

"Strong React + TS background. GraphQL gap is trainable given existing Node.js depth."



peaxis.app



AI Insight Panel

Screenshot placeholder

Insight panel in Candidate Drawer

Design rule: Every AI score must expose: numeric value · label (Strong/Good/Fair/Weak) · matched skills (teal chips) · missing skills (coral chips) · plain-language explanation. No silent black boxes.

Technical achievements

14

NestJS Feature Modules

auth, jobs, apps, AI, billing...

60+

REST API Endpoints

documented with Swagger

6

AI Features

parse · score · copilot · JD ·
cover letter · embedding

4

Microservices

Hire · Jobs · API · AI

3

Redis Cache Layers

embeddings 7d · CV 30d ·
results 24h

100%

Multi-tenant Isolation

every record scoped to
businessId

NOTEWORTHY IMPLEMENTATIONS

pgvector 1536-dim semantic search

Playwright PDF export pipeline

OAuth2: Google + LinkedIn

Stripe webhook billing

BullMQ background processor

Prisma schema-first migrations

Argon2 password hashing

Docker multi-stage builds

TypeScript

65%

Python

20%

SQL/Prisma

10%

CSS/Tailwind

5%

Product roadmap

Phase 1 — Core Platform (Done)

- ✓ NestJS API with 14 modules
- ✓ Multi-tenant auth + OAuth2
- ✓ Application pipeline (6 stages)
- ✓ Job CRUD + semantic search
- ✓ Stripe billing integration

Phase 2 — AI Layer (Done)

- ✓ FastAPI AI microservice
- ✓ CV parsing + match scoring
- ✓ Hiring Copilot + JD Generator
- ✓ BullMQ embedding pipeline
- ✓ Redis 3-layer caching

Phase 3 — Frontends (Done)

- ✓ PEAXIS Hire recruiter dashboard
- ✓ PEAXIS Jobs candidate portal
- ✓ Job Workspace with tabs + drawer
- ✓ Analytics dashboard
- ✓ Interview scheduling

Phase 4 — Future (Planned)

- Video interview integration
- Email notification system
- Candidate referral engine
- GDPR compliance toolkit
- Mobile-first candidate app

☒ Done ☐ In Progress ☐ Planned

Challenges & lessons learned

01

CHALLENGE

Multi-tenant data isolation at scale

Every query, guard, and background job must be scoped to a businessId without leaking data between tenants.

RESOLUTION

X-Business-ID header enforced at controller level. PrismaService always receives businessId from the guard. BullMQ jobs include businessId in payload.

KEY LESSON

Design security boundaries first — retrofitting scoping is 10x harder.

02

CHALLENGE

AI explainability vs. LLM cost

Deterministic scoring is cheap but limited. LLM scoring is rich but expensive and non-reproducible at scale.

RESOLUTION

Hybrid approach: deterministic skill-overlap for the score (reproducible, zero cost), Gemini Flash for the explanation text (low cost, high quality, cached 24h).

KEY LESSON

Separate "what" (deterministic) from "why" (LLM). Cache aggressively.

03

CHALLENGE

Frontend state vs. server freshness

Next.js server components cache aggressively. After inline edits (job form, stage moves), UI must reflect the updated state without full navigation.

RESOLUTION

Server actions call router.refresh() after mutations — triggers server component re-fetch without navigation. Client components own transient state only.

KEY LESSON

Understand Next.js cache invalidation deeply before building complex workspaces.

Future scope & roadmap



v2.0

Video Interview Integration

Embedded video screening with AI transcription, question generation, and candidate scoring.



v2.0

GDPR Compliance Toolkit

Candidate data deletion workflows, consent management, export requests, and audit logs.



v2.1

Candidate Referral Engine

Referral tracking, incentive management, and social sharing — turning every hire into a sourcing channel.



v3.0

Mobile-first Candidate App

React Native app for on-the-go job search, CV scan, and application tracking.

OPEN ITEMS & NEAR-TERM

- Email notification system (in progress)
- Elasticsearch full-text search (optional, configured)
- Advanced team collaboration (comments, @mentions on applications)
- API rate limiting per plan tier
- End-to-end test suite (Playwright)

Vision: PEAXIS becomes the OS layer for every hiring team — from sourcing to onboarding. The AI brain gets smarter with every hire, the candidate experience becomes the competitive advantage, and the data moat makes PEAXIS the default choice for growing companies.

PFE DEFENSE · ESPRIT · 2026

Thank you

Questions welcome

PROBLEM SOLVED	AI-powered hiring with full explainability + end-to-end platform
TECH STACK	Next.js · NestJS · FastAPI · PostgreSQL + pgvector · Redis · BullMQ · Gemini · Stripe
ARCHITECTURE	4 microservices · Multi-tenant SaaS · Queue-based AI pipeline · Cache-first strategy
DELIVERED	14 API modules · 60+ endpoints · 6 AI features · 4 Docker services · Production-ready

BIBANI Mohamed Habib Allah

École Supérieure Privée d'Ingénierie et de Technologie — ESPRIT
